A- SPECIFICATIONS OF MACHINES

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2/ Wiring diagram and working instructions.
3/ Construction.
4/ Main switch.
5/ Control wiring.
6/ Machine light.
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IMPORTANT

VPS CHECK LIST FOR EQUIPMENT QUALIFICATION TO BE FOLLOWED IN ALL RESPECT Ref: D / Meth / 027

A- SPECIFICATIONS OF MACHINES

It is obligatory on the part of the supplier to follow these specifications. Variations are only permitted with Valeo Friction Materials India's prior approval in writing.

1. VOLTAGE AND TYPE OF CURRENT

The electrical equipment shall be suitable for supply voltage of 415 # 5 % V between phases and 240 V between any phases and neutral at a frequency of 50 cycles.

1.1 DC controls in relay technique 24 V.

1.2 Pneumatic valves: for AC control current circuits 220 V
For AC control current circuits 24 V
Make: Festo/SMC only

1.3 Cable make: Finolex

2. WIRING DIAGRAM AND WORKING INSTRUCTIONS

Every machine has to be delivered with:

2.1 Complete set of transparent drawings showing the power circuits, control circuit diagram wiring, diagram and a parts list containing the designations of the electrical and other parts connected with the electrical system.

2.2 A general plan, which shows the position of the various electrical parts (layout plan of electrical parts).

2.3 Description of the mechanical and electrical function of the machine (sequence of operation).

2.4 In respect of memory programmable controls, the following are to be supplied: electrical symbol list, programme lists and prints of circuit plan with symbolic addresses. The type of data carrier must be agreed to be with the purchaser, such as the control programme on data carriers such as cassettes, punched strips or floppy disc.

2.5 In G5 Mixing stn Vibrator should function only with load ,such way it should program (Hopper and Discharge Pipe)

3. CONSTRUCTION

3.1 Control Equipment

Control equipment (contractors, fuses, overload relays, control transformers, etc.) shall be grouped together, easily accessible, housed in one completely enclosed space inside the machine or in a separate control cabinet.

1. Panel consisting of up to 3 contactors and 1 timer to be integrated with operators'

panel.
2. Circuit requiring more than 5 contactors should use a PLC.
3. The panel should be dust, moisture and vermin proof.
4. Rubber beading should be provided on the periphery of the door.
5. Width of panel door should not exceed 0.5 metres.
6. Provide earthling PVG on exterior and earthling strip on interior.
7. Outgoing cables to be taken from bottom or top of panel through suitable ducts.
8. All components to be mounted on Bakelite sheet of at least 6 mm thickness.
9. All similar components to be mounted in a row on DIN rails. A distance of 8 mm to be maintained between each component.
10. All panels to be front wired and to be provided with PVC channels between each row of components and along the edges of the bakelite sheet.
11. Distance of 15 mm to be maintained between channel and element for easy maintenance.
12. Numbering of wires to be unique. All wires terminating at the same node should have the same number.
13. Extra 20% of number of wires passing through conduits are to be kept as spare.
14. Where numbers of outgoing cables are more than 4 suitably designed cable trays are to be used.
15. Cable tray to be terminated to a junction box and wiring to be further distributed to individual elements.
16. Junction box to be so located and designed as not to obstruct operator / maintenance personnel or encourage unsafe practices.
17. Door to be provided with panel lock on the junction box.
18. All cable going to machine elements like motors, solenoid valves etc., to be clamped and bound properly so as not to cause injury to personnel or obstruction to machine movements.
19. Make: KRYCARD energy meter should be provided for energy monitoring

3.2 Operating buttons and switches

3.2.1 The push buttons and switches which are required to be operated frequently shall be placed in front of the machine at a height of 1050.

3.2.2 Push button switches and other switches, which are operated occasionally, need not be close to the operator.

3.2.3 Emergency off push-button shall be a red mushroom type and be placed within the easy reach of the operator with bright yellow sticker

3.3 Push Buttons

3.3.1 Colour of push buttons :
The applications of colours for push buttons used on machine tools / equipment shall be placed as indicated in the table below :

Colour	***Function***	***Example of Application***
Red	Stop	- Stop one or several motors

		- Stop of machine element - De-energising of magnetic chuck - Stop of the cycle (if the operator pushes button, the machines stops after relevant cycle is completed).
Yellow	Starting of a return motion not in the usual operating sequence. Or start of an operation intended to avoid dangerous condition.	Return of machine elements to the starting point if the cycle has not been completed. Press the yellow push-button may over-ride functions which have been selected previously.
Green	Start (preparation)	- Energising of the control circuits. - Start of one or several motors for auxiliary functions. - Start of machine elements. - Energising of magnetic chucks.
Green or black	Start (execution)	- Start of a cycle or a partial sequence - Inching - Jogging
Light blue	Any function not covered by the above colours	- Control of auxiliary functions which are not directly related to the working cycle. - Re-set of protective relays (if the same button is used for « stop » it shall be red).

3.4 **Marking of push button**

All the relevant functions of the push buttons shall be marked with inscribed anodised or engraved labels.

In case of push button control for one motor only, « O » and « I » can also be used.

3.5 **Mushroom head push button**

Mushroom head « red » push button shall be exclusively used for the emergency stop both in automatic and in manual operation. The push button should be so provided that it completely stops the machine operation and cuts off power to the entire circuit when it is actuated.

Mushroom head « black » button shall be used for start of cycle buttons for operation with both hands.

The signal indicating lamp lens colours for the signalling lamps used on machine tool equipment shall be as indicated in the table below :

Colour	*Significance / Indication*	*Example of Application*
Red	Abnormal conditions requiring immediate action by the operator	Order to stop the machine immediately (e.g. because of an overload) OR To indicate that protective device has stopped the machine (e.g. because of an overload, over-travel or another failure).
Yellow (Amber)	Attention or caution	Some value (current, temperature) is approaching its permissible limit OR automatic cycle running).
Green	Machine ready	Machine ready for operation. All necessary auxiliary functioning units in starting position and hydraulic pressure or output voltage of a motor generator in the specified range, etc. Cycle completed and machine ready to be re-started.
White	Circuit energised normal condition	Main switch on « On » position - Choice of the speed or the direction of rotation - Auxiliaries not related to the working cycle are functioning
Blue	Any significance not covered by the above colours	- Selector switch in setting position - A unit in forward position - Micro-feed of a carriage or unit

3.6 Limit Switches

Limit switches are to be properly marked as per the wiring diagram and shall be mounted in an easily accessible position. They shall be well protected against dirt, dust and liquids.
Double Pole (2NO + 2NC) element –BCH/Omron make to be used for long life

3.7 Junction Box

Wherever junction of wires are made, they shall be terminated in terminal strips housed in a completely enclosed junction box of ample size and with proper ear thing
Cable gland to be provided for Proper Control cable support.

3.8 Protection

3.8.1 *Housings*

The construction of control cabinet, control station, junction box etc., shall be such that dust, coolant, lubricating oil and metal chips shall not enter this portion and also it shall be easily accessible for maintenance. The control cabinet and junction boxes shall be opened only by a special key / tool.

3.8.2 *Cable Channel*

It is recommended that the dimensions of conduits and raceways should permit the easy laying of additional cables in addition to the power cables at a later date if required.

3.8.3 *Conduits*

While choosing the position and laying of the conduits for the various electrical items, the possibility of mechanical damage and effects of chips and coolant liquids have to be taken into consideration. All the conductors (wires and cables) shall be protected mechanically strong casing throughout their length. Finolex PVC conduit reinforced shall be clamped and protected against abrasion and deterioration.

3.9 **Terminals**

Terminals, which are still live when the main switch is switched off, shall be protected properly so that maintenance personnel do not come in contact during repairs. In addition, a clear indication shall be given on the protective cover of such contacts that they are live.

At the points where mains will have to be connected, there shall be an additional insulated terminal for the neutral. The doors and covers of control cabinet, junction boxes etc. are to be provided with a flashing light in red colour as per IS : 1356 - 1964.

3.10 Wherever valves, limit switch etc. are grouped together, they shall be wired independently and not through each other.

3.11 **Motors for coolant pumps**

Separate plug-in-sockets should be provided for connecting the coolant pump motor to the machines.

4. **MAIN SWITCH**

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A hand operated main isolating switch to withstand full load capacity of the machine is to be provided in the control cabinet. It shall be so built that an accidental switching-in is not possible.

5. CONTROL WIRING

5.1 Wherever a machine has more than one motor (valves, clutches, heaters, etc. or other power consumer), the control wiring shall be through control transformer.

5.2 Control transformer

The control transformer shall have primary and secondary windings, which are separate from one another. It shall be of adequate rating primary winding for 415 V + 5% tapping. The secondary of the control transformer shall be protected with two-delayed type cartridge fuses and secondary with separate fuses for control circuit, magnetic valve circuit, machine light socket, etc.

6. MACHINE LIGHT

6.1 Machine light shall be connected through control transformers.

6.2 Where there is no control transformer, the machine lights shall be wired between phase and neutral with sufficient protection

6.3 The fuse for the machine light shall not be rated more than 2A and shall have a separate switch.

6.4 The machine light shall be earthen separately.

7. MOTORS

3 PHASE AC induction squirrel cage motor, unless otherwise specified. The motor shall ne NGEF / SIEMENS / KIROLOSKAR / HINDUSTAN BROWN BOVER make.

7.2 Type of Switching and Voltage

Stator winding shall be suitable for direct switching, unless otherwise specified. The insulation of the windings shall be class E, unless otherwise mentioned.

7.3 Motors including the terminal boxes must be easily accessible for maintenance jobs and protected against entry of dust / dirt as well as reactions from fluids / vapours. The flow of cooling air shall not be hindered.

The specified direction of rotation must be marked on the motor clearly.

7.4 Protection

7.4.1 All motors shall have on all the 3 phases thermal over-current protection of the hand re-setting type as well as short circuit protection by means of delayed action fuses. « No Voltage » protection is necessary from the safety point of view.

- The motor shall be of totally enclosed execution, unless otherwise specified.
- Motor should be earthed with proper size copper wire

7.5 Name Plate

When the name plate on the motor cannot be seen in the mounted condition, a second name plate is to be mounted additionally on the machine tool in a suitable place.

8. SWITCH GEAR

Switch gear shall be preferably of Siemens India make.
Exceptions only with our previous approval.

9. EARTHLING OF METAL PARTS

All metal parts of the machine-tool accessories in which electrical equipment is contained, whether an integral part of the machine-tool itself, whether moving or independently mounted, shall be connected together electrically so that the entire machine-tools can be connected to earth.

The resistance measured between the main earthling terminal and any of the machine-tool associated metal parts which may become live due to an insulation fault shall not exceed 0.1 ohm.

If multiple conductor cables are used for connection to moving parts or pendant stations, an earth continuity conductor shall be provided in the cable.

Screws and terminals for the connection of earth continuity conductors shall not have any additional function of mechanical fastening.

9.2 Main Earthling Terminal

A main earthling terminal shall be provided close to the main input terminals. It shall be of such a size as to enable the connection of an earth continuity conductor of the following cross section.

Cross-section of the main conductors supply the equipment	Cross-section for which the main earthling terminal has to be dimensioned
Up to and including 16 mm²	Equal to that of the main conductors

Larger than 16 mm²	At least 50% that to the main conductors with a minimum of 16 mm²

10. WIRES

Flexible wires shall be preferred for control circuits.
Single or standard wires shall be preferred for power circuits.

10.1 Cross Sections

Voltage Grade	Minimum Cross Sectional Area (sq. mm)	Type of Conductor	IS Reference
650 V / 1100 V	1.0	Standard wires of copper for control circuits	IS : 694 (Part 1 and II) 1964

* for PLC controls 0.5 mm² (min) cross section for control cables are acceptable.

Colour Code as per IEC 2C 4.1/1965 (with charges)

Colour	*Applications*
Black	3 phase AC / DC power circuit
Light Blue	Neutral (MF) power circuit
Green with yellow stripes	Protective earth
Red	AC control circuit
Blue	DC control circuit

11. SAFETY PROTECTION

In case of two hand safety operations with two push buttons, the wiring system shall be so arranged that after previous operations are fully realised, the new switching sequence can follow only when both the push buttons are released and actuated again. Also the circuit / programme should be such that the cycle can operate only when the 2 push buttons are actuated within 0.5 SECs of each other (anti-tie circuit).

12. MARKINGS AND LABELLING

12.1 All electrical equipment, contractors, switching elements, terminals etc. are to be labelled as per circuit diagram.

12.2 All wires are to be identified with suitable cable markers according to numbers or letters on terminals.

12.3 **Name Plate**

On every control cabinet there shall be a name plate which will give electrical date.

Machine	
Electrical circuit diagram number :	
Conn. load :	kW / HP
Supply voltage :	V
Control voltage :	V
Frequency :	HS
Current :	A
Main fuse :	A
Manufacturer's name and address :	

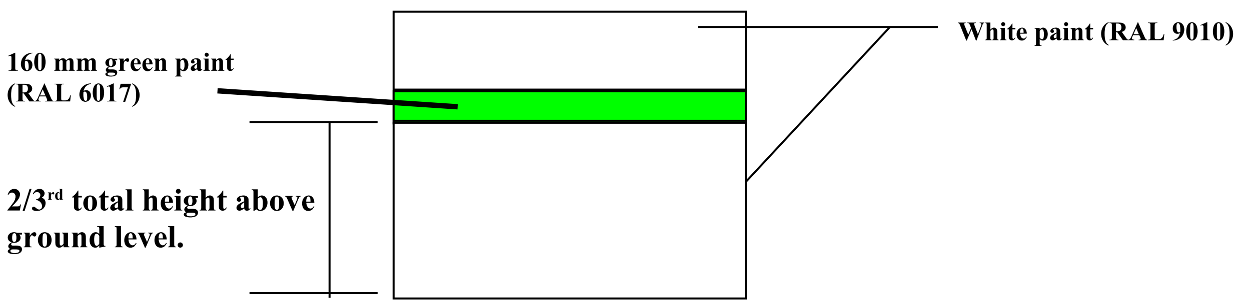
12.4 Machine should be provided with cycle counter

List of preferred makes of Trade Standard items to be used on our machinery / equipment.

- **Painting**

The Epoxy white paint (RAL9010) shall be applied to Machine, Electrical Panels (outside & inside) Hydraulic units, and other accessories.

The green strip paint (RAL 6017) shall be applied to machines, Electrical Panels and other big prominent accessories if any. The strip width being 160mm the bottom of the strip at 2/3rd of total M/c height.



I. Mechanical

1.	Machine spindle bearings Precision bearings, P5, P6 grade	SKF, FAG, FAFNIR, TIMKEN RHP
2.	Bearings for internal grinding simple	RHP / FAG
3.	Bearings for general application	As per S. n° 1 above + Koyo Norma
4.	Needle bearings	INA / NTN / FAG / IKO
5.	V belts (non standard sizes to sizes to be avoided)	Fenner (India) Ltd.
6.	Flat belts (rubberised woven endless cotton belts) - For light applications - For medium applications	 Fenner India
7.	Poly « V » Belts	Dunlop/Fenner/Poly fix
8.	Polyflex belts	Dunlop/Fenner/Poly fix
9.	PIV belts - Chains - Belts	 Dunlop/Fenner/Poly fix
10.	Chains	Rolon, Diamond
11.	Electro-magnetic clutches	EMCO Lenze
12.	Couplings	Lovejoy
13.	Pulleys	Fenner
14.	Oil seals and « O » rings	Vaco
15.	Bellows	Sur-Henning, Vaco
16.	Slide wiper	Sur-Henning, Raksha
17.	Geared motors	New Allenbury, Shanti
18.	Reduction units	Radicon, Elecon
19.	Disc springs	

20.	Toothed belts (timing)	Uniroyal
21.	Conveyor belts (endless)	NTB
22.	Liner slides bearings	THK Japan
23.	Fasteners	Unbrako, Sundaram Fastener

II. **Hydraulic** (only metric sizes)

1.	Pumps (Variable Displacement Vane Pumps Gear)	 Vickers, Rexroth, Yuken Dowtry, Vickers, Rexroth
2.	Valves	Vickers, Rexroth, Bosch
3.	Pressure switches	Rexroth, Danfoss
4.	Pressure gauge isolator	Fenner, Hydax
5.	Return line filters	Vickers, Hydax, Hydroline
6.	High pressure hydraulic filter	Hydax
7.	Hydraulic motor	Rexroth, Vickers
8.	Accumulator	Bosch, Vickers, Rexroth
9.	Pipe joints / fitting	Hydair, Kim
10.	Flexible hoses	Dunlop, Swastik
11.	Cylinders	Integral Automation
12.	Pressure Gauge (glycerine filled)	Fiebig, Gluck,Waree
13.	Manometer precision connectors / fittings	 Hydro-Technic
14.	Float switches	Mini-Hyd
15.	Quick Release Couplings	
16.	Hydraulic oil	Hydrol – 68 (BP)
17.	Mainram seal Type	Polyurathene

III. **Pneumatic** (only metric sizes)

1.	Cylinders (only metric)	Festo, SMC
2.	Valves - Solenoid actuated - Non solenoid	Festo,SMC Dancal
3.	**FRL Units with Digital Pressure Switch**	Festo,SMC
4.	Pressure switches	Danfoss (Indfoss)
5.	Pressure gauges	FEST,Waree, Fiebig, Cluck
6.	Nylon tubings	Festo,SMC
7.	Fittings	Festo,SMC
8.	Ball valves (hand operated)	Audo (SS),Italy make
9.	Quick release couplings	Festo,SMC
10.	Air guns	Shavo Norgren
11.	Filters	Shavo Norgren
12.	Air purging valve	ASCo make
13.	Manifold valves	SMC,FESTO
14	Box for Pneumatic Valves	To avoid sol.coil wire cut during Machine cleaning

IV. **Refrigeration & Air Conditioning**

1.	Compressor	Copeland- Kirolosher
2.	Expansion valve	Danfoss
3.	DC solenoid valve	Danfoss
4.	Thermostat	Ranutron, Danfoss, Cuttler
5.	Pressure switch	Hammer

6.	Stop valve	Danfoss
7.	Freon dryer	Bap, Danfoss
8.	Manometer for Freon	Fiebig
9.	Refrigerant gas	Freon -22

V. **Coolant units**

1.	Multistage pump	
2.	Self priming pump	Kirloskar Brothers
3.	Centrifugal pump	Rajamane
4.	Coolant valves	Avcon
5.	Couplings	Hydax
6.	Chemical pump	Vulcan lavel (S.S.)
7.	Direction control valve	Avcon

VI. **Lubrication units**

Centralized lube unit ,Cen lub make

VII. **Electrical**

1.	Limit switches (oil tight)	BCH, Jaibalaji
2.	Micro switches	Kaycee
3.	Relays / Contractors, Overload relays	Siemens, BCH
4.	Fuses	English Electric, Siemens, L&T
5.	Motors AC, 3 Phase	NGEF / Bharat Bijlee Kirloskar Electric
6.	Push buttons	Tecknic, Siemens

7.	Brake motors	Kirloskar Electric
8.	Brake units added for motor	Cuttler Hammer
9.	Timers - Electronic	Siemens
10.	Main switches	Siemens, L&T
11.	Selector switches	L&T, Siemens
12.	Miniature relays	OEN, PLA
13.	Terminal connectors	Elmex
14.	a)Meters b) Energy Meter	
15.	Digital temperature indicating controllers	Autonix
16.	Temperature recorders	Yokogawa
17.	Wires / cables	Power cable- Polycab/Orbit Control Cable -Finolex Heater Cable-Fibre Glass cables
18.	DC Motor	Siemens/ABB/Kirlosker
19.	Single phase FHP motors	Remi, Tuku
20.	Multi plunger limit switches	Teknic Euchner, Balluff
21.	Cycle counters	Gujral
22.	Pre-set counters with relay	Transducers & Allied Products
23.	Foot switches	Cuttler Hammer
24.	Panel with glass doors where ever possible for TPM	
25.	Digital (above 10HP) Motor protection relay – Telemechanique	
26.	Seq .Timer	EAPL make
27.	Thermocouple	Metal briaded common setting cable

VIII. **Electronics**

1.	PLCs	Siemens/Mitsubishi/ABB
2.	CNCs	Fanuc
3.	Proximity sensors	Transducers & Allied Products SNA Electronics, Balluff/TAP/IFM
4.	HMI	Proface
5.	Drive	Schinider/Danfoss

B- STANDARDS FOR MAINTAINABILITY - HYDRAULIC

1. Hydraulic Pimps : always use variable displacement pumps (unless otherwise specified or agreed upon).

2. All connections / piping of the machines to be done with seamless pipes of suitable size.

3. Use manifold blocks irrtead of pipes.

4. All elements in a power pack should be numbered.

5. A laminated hydraulic circuit diagram should be provided on the hydraulic power pack.

6. All solenoid valves should have LED indication of actuation.

7. All connections between machine and power pack should be through flexible hoses of suitable size.

8. A tray should be provided below the power pack to collect and contain leakage.

9. Hydraulic connections to fixtures etc. should be through quick connect couplings.

10. Power pack should contain clogging indicators on filters.
11. A laminated hard copy of hydraulic circuit to be provided in the panel
12. Air purging pneumatic lines should be supported with proper MECH arrangements

C- STANDARDS FOR MAINTAINABILITY - GENERAL

1. Machines checking mandrels and tool setting gauges to be provided with the machine.

2. Quick set-up change devices like quick-change tooling / fix truing to be incorporated in all machines.

3. Machines shall be treated with I coat of anti rust coating and then painted with 2 coats of colour.

4. The following documents are to be provided along with the machine.

   - Operating and maintenance manual. (3 sets)
   - List & drawings spare parts / wear out parts. (3 sets)
   - All circuits - Electrical / Hydraulic / Pneumatic. (3 sets)
     (Whichever applicable)
   - List / specification / sources of brought out items. (3 sets)
   - Machine FMEA (3 sets)
   - Recommended spares to stock for next 2 years
   - Soft copy and hard copy of drawings (1set)
   - Laminated hard copy of electrical diagram should be provided in the panel